Stress Assignment without Morphology in Portuguese: the qualityless vowel /0/ and the Prosody-Conditioned Extrametricaliser (PCE)[1]

Shinya Makino
itakanro@g.ecc.u-tokyo.ac.jp

Abstract:

This paper offers an Optimality-Theoretic reinterpretation and synthesis of the author's earlier Japanese works (Makino 2010, 2012) on Portuguese stress patterns, focusing exclusively on the theoretical essentials. We put forward a **prosody-only account of Portuguese word stress**. The analysis reduces the Portuguese stress system to the **interaction of:**

- **Qualityless vowel /0/ bearing one mora**
- **Prosody-conditioned extrametricaliser (PCEs)**
- **General metrical theory formalised through OT constraints**

This /0/ is the musical "**ghost note**" — that is, a **beat** (moraic temporal position) without **timbre** (segmental quality) — which nevertheless shapes the **rhythm** (strong–weak alternations), as exemplified by /0/ in *contador* /kont-a-dor-**0**/ (cf. "**full-note**" /a/ in *contadora* /kont-a-dor-**a**/). PCEs constitute a class of morpheme-sized items that, when **the prosodic condition is satisfied, render the final syllable of the prosodic word extrametrical**. The principles and parameter of primary stress assignment in Portuguese are as follows:

Principles

- Trochaic binary footing from the right edge
- The rightmost foot functions as the head foot

Parameter: determination of the right edge

- **Value 1:** final syllable — unmarked default
- **Value 2:** penultimate syllable — marked (when a PCE satisfies the condition)

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Metrification is **computed strictly from phonological information**, including a single piece of **non-morphosyntactic-categorial, lexicalised phonological information** — namely, **the PCE condition**. No reference to syntactic categories or to morphological bracketing is made at any point in the stress-assignment process. This approach thus **eliminates morphology from stress computation proper** and unifies items such as *contador*, *contadora*, *falava*, *falávamos*, *faláveis*, *número*, *numeroso*, *numérico*, *sobre* and *após* within a single minimal mechanism.

0. Abbreviations, symbols and diacritics

PrWd = prosodic word; **Ft** = foot; **HdFt** = head foot; **PCE** = *prosody-conditioned extrametricaliser*; **σ_{final}_PCE** = the syllable whose nucleus is the final vowel of a PCE; **L** = light syllable; **H** = heavy syllable

FTBIN(σ) = Parse syllables into a binary foot; **TROCHEE(σ)** = Parse syllables into a trochaic foot, where the first syllable is strong and the remaining syllable(s) are weak; **PARSE(σ)** = Parse every syllable into a foot; **ALIGN-R(Ft, PrWd)** = Align the right edge of each foot with the right edge of the prosodic word; **ALIGN-R(HdFt, PrWd)** = Align the right edge of the head foot with the right edge of the prosodic word; **ALIGN-R(Ft, σ_{final}_PCE)** = Align the right edge of each foot with the syllable whose nucleus is the final vowel of a PCE; **ALIGN-R(HdFt, σ_{final}_PCE)** = Align the right edge of the head foot with the syllable whose nucleus is the final vowel of a PCE; ***CLASH** = Avoid a stress clash; **HDFTBIN(σ)** = Require the head foot should consist of two syllables

/ / = underlying representation; **[ ]** = surface representation; → = derivation; **-** = morpheme boundary; **#** = word boundary; **/0/** = qualityless vowel bearing one mora; **C** = consonant; **C̩** (e.g. r̩, n̩, s̩) = syllabic consonant that arises from the association of a consonant with the mora supplied by /0/; **μ_0** = the mora supplied by /0/; **.** = syllable boundary; ˈ = primary stress; ˌ = secondary stress; **σ** = syllable; ( ) = footing

1. Introduction

While many previous accounts rely on morphological information in stress computation, the present approach **dispenses with morphology**. Building on the prosody-based insight of Mateus & d'Andrade (2000), while adopting a different formal implementation, it **derives stress solely from phonological information** — namely, **the syllable string, syllable count and the PCE condition**.

2. The qualityless vowel /0/

In the model proposed in this paper, a **qualityless vowel /0/ that bears one mora** in underlying representations is postulated — e.g. *contador* /kont-a-dor-**0**/ (cf. *contadora* /kont-a-dor-**a**/); *jacaré* /ʒakarɛ-**0**/; *trabalhar* /trabaʎ-a-r**0**/; *trabalharmos* /trabaʎ-a-r**0**-mos/; *após* /a-pɔ**0**s/; *até* /atɛ**0**/. This vowel /0/ is the musical "**ghost note**" — **beat** (moraic temporal position) without **timbre** (segmental quality) — which nevertheless shapes the **rhythm** (stress pattern: strong–weak alternations). It is a purely phonological entity, serving various morphological purposes: in *contador* /kont-a-dor-**0/** and *jacaré* /ʒakarɛ-**0/**, the /0/ functions as the thematic vowel — stem vowel or *índice temático* (stem marker) — just as /a/ does in *contadora* /kont-a-dor-**a/** (also analisable as /kont-a-dor-0-**a/**, with /0/ deleted when syllabified). On the other hand, it forms part of the infinitive or future subjunctive suffix /-r0/ in *trabalhar* /trabaʎ-a-r**0/**, and likewise occurs in the preposition *até* /atɛ**0/**, which does not display any internal morphological structure.

The postulation of /0/ is not an ad hoc assumption but rather a corollary deduced from a stress theory based on the metrical foot. Strong–weak alternation presupposes at least two distinct temporal positions on the time axis. Without such positions, alternation in strength cannot be realised. Conversely, distinct temporal positions do not presuppose strength. It follows that temporal positions can exist independently of strength. In other words, **distinct temporal positions are a prerequisite for strong–weak alternation**, but not vice versa.

If, metrical-theoretically, a stress pattern is defined as the *iterative repetition of the alternation between strong and weak beats along the time axis*, then both strong and weak beats presuppose distinct temporal positions. When the main stress — the most prominent strong–weak alternation — falls on *-dor* in *conta**dor***, **there must necessarily exist within *-dor* the temporal position corresponding to the weak beat paired with the strong one**. If such beats are borne by the syllable — more precisely, by the syllable nucleus constituted by a given moraic temporal position and associated with some segmental quality — then, since the strong beat is borne by the syllable nucleus /o/ as one moraic temporal position, **the paired weak beat must be borne by the syllabic mora /0/ as the other moraic temporal position**. Thus, positing /0/ — a qualityless, syllable-nucleus-projecting mora — is required both by **the primacy of temporal positions over strong–weak alternation** and by **the very definition of the stress pattern**.

Accordingly, if /0/ were not assumed, *-dor* would constitute a degenerate foot — which, within metrical theory, is by definition a monosyllable left over at the end of footing — and this imperfect foot would serve as the head of the prosodic word, bearing the highest prominence. This outcome would be theoretically undesirable, since footing in Portuguese is assumed to proceed from the right edge, as shown in Sections 4–6, and yet its initial foot would paradoxically be degenerate — a state of affairs for which metrical theory may offer no principled explanation. For further support, **empirical corroborations** will be presented in Section 10.

It should be noted, finally, that prior to the present study, related ideas had already been proposed in Mateus & d'Andrade (2000) and Mateus *et al.* (2003). These works attempt to account for Portuguese stress pattern without recourse to constituents such as metrical feet, analysing it instead in terms of a "rhythmic wave" in which troughs and peaks of prominence alternate (Mateus & d'Andrade 2000: 122). For instance, in *contadora*, the rhythmic wave is assumed to begin at the word-final syllable *-ra*, taken as the starting position, and to propagate from right to left, beginning with a trough; the first peak reached along this wave, *-do-*, receives the strongest prominence. As for words such as *café*, they are assumed to contain a "defective class marker" which, "though phonetically null, has a rhythmic position" (Mateus and Andrade 2000: 123); the rhythmic wave is thus taken to originate at this phonetically null rhythmic position. T he differences between their approach and the present study are as follows. First, their analysis employs a skeletal tier rather than a moraic tier. Second, the phonetically null rhythmic position is associated with a specific morpheme — the defective class marker characteristic of nouns and adjectives. In order to account for the penultimate stress of infinitives and future subjunctive forms such as *trabalhar* [ˌtrɐ.βɐˈʎaːɾ], this phonetically null rhythmic position would necessarily have to be assigned to the infinitival and future subjunctive ending *-r*, yet no discussion of this issue is provided. It also remains unclear how this notion could be extended to words such as the adverb *até*, which do not display any internal morphological structure.

3. The prosody-conditioned extrametricaliser (PCE)

It is also postulated that there exists a class of ***prosody-conditioned extrametricalisers*** **(PCEs)**. These are **morpheme-sized items** that are **lexically specified to render the final syllable of the PrWd extrametrical**, but **only when their own final vowel serves as the nucleus of the penultimate syllable of the PrWd** — e.g. *-ic-* /-ik-/ in *acadé*m***ic****o* /akadɛm-**ik**-o/; *-vel* /-vel/ in *agradá**vel*** /a-grad-a-**vel**-0/; *numer-* /numɛr-/ in *número* **/numɛr**-o/; *-va-* /-va-/ in *trabalhá**va**mos* /trabaʎ-a-**va**-mos/; *-r-* /-r0-/ in *trabalha**r**des* /trabaʎ-a-**r0**-des/. **The membership of a morpheme in this class is independent of its morphosyntactic category**.

4. Unmarked metrification

Regardless of syntactic category or morphological structure, **iterative formation of trochaic binary feet proceeds from the final syllable of the PrWd towards the word-initial edge**, with the **rightmost foot** serving as the head and bearing **primary stress**:

possibilidade /pɔs-i-bil-i-dad-e/ → pɔ.si.bi.li.da.de → (ˌpɔ.si)(ˌbi.li)(ˈda.de) → [ˌpu.siˌβɨ.liˈðaː.ðɨ]
contador /kont-a-dor-0/ → kon.ta.do.r̩ → (ˌkon.ta)(ˈdo.r̩) → [ˌkõ.tɐˈðoːɾ]
contadora /kont-a-dor-a/ → kon.ta.do.ra → (ˌkon.ta)(ˈdo.ra) → [ˌkõ.tɐˈðoː.ɾɐ]
coração /korason-0/ → ko.ra.so.n̩ → (ˌko.ra)(ˈso.n̩) → [ˌku.ɾɐˈsɐ̃ũ̯]
parabéns /para-ben-0-s/ → pa.ra.ben.0s → pa.ra.ben.es → (ˌpa.ra)(ˈben.es) → [ˌpɐ.ɾɐˈβɐ̃ĩ̯ʃ]
jacaré /ʒakarɛ-0/ → ʒa.ka.rɛ.ɛ → (ˌʒa.ka)(ˈrɛ.ɛ) → [ˌʒɐ.kɐˈɾɛː]
trabalhamos /trabaʎ-a-mos/ → tra.ba.ʎa.mos → (ˌtra.ba)(ˈʎa.mos) → [ˌtrɐ.βɐˈʎɐː.muʃ]

trabalhais /trabaʎ-a-des/ → tra.ba.ʎa.des → (ˌtra.ba)(ˈʎa.des) → [ˌtrɐ.βɐˈʎai̯ʃ]

trabalhar /trabaʎ-a-r0/ → tra.ba.ʎa.r̩ → (ˌtra.ba)(ˈʎa.r̩) → [ˌtrɐ.βɐˈʎaːɾ]

trabalhei /trabaʎ-a-i/ → tra.ba.ʎa.i → (ˌtra.ba)(ˈʎa.i) → [ˌtrɐ.βɐˈʎɐi̯]

trabalhou /trabaʎ-a-u/ → tra.ba.ʎa.u → (ˌtra.ba)(ˈʎa.u) → [ˌtrɐ.βɐˈʎoː]

sobre /sobre/ → so.bre → (ˈso.bre) → [ˈsoː.βɾɨ]

This suggests the following constraints in interaction: **FTBIN(σ); TROCHEE(σ); PARSE(σ); ALIGN-R(Ft, PrWd); ALIGN-R(HdFt, PrWd)**. Note that the syllabification of /0/ is hypothetical and that various possibilities are theoretically conceivable. What is crucial from a metrical-theoretical perspective, however, is that **/0/ is not merely a so-called stray/floating mora**, but **rather an element projecting a syllable nucleus**, and therefore that the syllable containing the mora supplied by /0/ — henceforth μ_0 — is a separate syllable from those preceding and following it. **The notation /0/** is adopted precisely to indicate that it constitutes **a syllabic unit**. Note also that, whereas the syllabification of segments is assumed to refer not only to phonological but also to morphological information, the metrification of syllable strings is assumed to refer solely to phonological information.

5. Marked metrification

If the **final vowel of a PCE** serves as the **nucleus of the penultimate syllable of the PrWd** (the **PCE condition**), the same iterative formation of trochaic binary feet proceeds **from that penultimate syllable** — **rather than from the final one** — with the **rightmost foot** serving as the head foot and bearing **primary stress**. The final syllable is thereby excluded from metrification — that is, *extrametricalised* — and remains unstressed:

*académ**ico*** /akadɛm-***i*k**-o/ → a.ka.dɛ.m***i*.k**o → (ˌa.ka)(ˈdɛ.m***i*)k**o → [ˌɐ.kɐˈðɛː.mi.ku]

*agradá**vel*** /a-grad-a-**v*e*l**-0/ → a.gra.da.**v*e*.l̩** → (ˌa.gra)(ˈda.**v*e*)l̩** → [ˌɐ.ɣɾɐˈðaː.vɛɫ]

númer**o* /**num*ɛ*r**-o/ → **nu.m*ɛ*.r**o → (ˈ**nu.m*ɛ)**r**o → [ˈnuː.mɨ.ɾu]

túnel /**tun*e*l**-0/ → **tu.n*e*.l̩** → (ˈ**tu.n*e***)**l̩** → [ˈtuː.nɛɫ]

fémur**es* /**fɛm*u*r**-0-s/ → **fɛ.m*u.r0s → **fɛ.m*u***.res → (ˈ**fɛ.m*u***)res → [ˈfɛː.mu.ɾɨʃ]

***sótã**o* /**sɔt*a*n**-o/ → **sɔ.t*a*n**.o → (ˈ**sɔ.t*a*n**)o → [ˈsɔː.tɐ̃ũ̯]

*trabalhá**va**mos* /trabaʎ-a-**v*a***-mos/ → tra.ba.ʎa.**v*a***.mos → (ˌtra.ba)(ˈʎa.**v*a***)mos
→ [ˌtrɐ.βɐˈʎaː.vɐ.muʃ]

*trabalhá**ve**is* /trabaʎ-a-**v*a***-des/ → tra.ba.ʎa.**v*a***.des → (ˌtra.ba)(ˈʎa.**v*a***)des → [ˌtrɐ.βɐˈʎaː.vɐi̯ʃ]

*trabalha**r**des* /trabaʎ-a-**r*0***-des/ → tra.ba.ʎa.***r̩***.des → (ˌtra.ba)(ˈʎa.***r̩***)des → [ˌtrɐ.βɐˈʎaɾ.ðɨʃ]

By contrast, if the **final vowel of a PCE** does **not function as the nucleus of the penultimate syllable of the PrWd**, **unmarked metrification**, as described in Section 4, **emerges**:

*atoni**c**idade* /a-tɔn-***i*****k**-i-dad-e/ → a.tɔ.n***i*.s**i.da.de → (ˌa.tɔ)(ˌn***i*.s**i)(ˈda.de) → [ˌɐ.tuˌni.siˈðaː.ðɨ]
***numer**oso* /**numɛr**-oz-o/ → **nu.mɛ.r**o.zo → (ˌ**nu.mɛ**)(ˈ**r**o.zo) → [ˌnu.mɨˈɾoː.zu]
*trabalha**va*** /trabaʎ-a-**v*a***/ → tra.ba.ʎa.**v*a*** → (ˌtra.ba)(ˈʎa.**v*a***) → [ˌtɾɐ.βɐˈʎaː.vɐ]
*trabalha**r*** /trabaʎ-a-**r*0***/ → tra.ba.ʎa.***r̩*** → (ˌtra.ba)(ˈʎa.***r̩***) → [ˌtɾɐ.βɐˈʎaːɾ]

When a PrWd contains **more than one PCE**, metrification proceeds from **the syllable whose nucleus is the final vowel of the PCE that satisfies the PCE condition** stated above:

numéric**o* /**numɛr-*i*k**-o/ → **nu**.**mɛ**.**r*i.**k**o → **nu**(ˈ**mɛ**.**r*i***)**k**o → [nuˈmɛː.ɾi.ku]

This phenomenon, in which items such as *-ic-*, *numer-*, *-va-*, *-r* shift leftward or rightward and thereby become inactive, indicates that a PCE is a **purely prosodic/metrical unit**: it **triggers extrametricalisation only under specific prosodic conditions**. This constitutes a binary prosodic on/off mechanism, **clearly distinct from general lexical accent specification**. All of the above suggests the ranking: **ALIGN-R(Ft, σ_{final}_PCE) ≫ ALIGN-R(Ft, PrWd)** and **ALIGN-R(HdFt, σ_{final}_PCE) ≫ ALIGN-R(HdFt, PrWd)**.
In the rhythmic wave theory discussed in the last paragraph of Section 2, the morpheme *-ic-*, as in *atónico*, is treated as a "lexically pre-assigned trough", from which the rhythmic wave is assumed to begin and to propagate from right to left (Mateus & d'Andrade 2000: 124). This assumption successfully accounts for the stress pattern of *atónico* as [ɐˈtɔː.ni.ku]. In the morphologically related noun *atonicidade*, which contains the same morpheme *-ic-*, however, the rhythmic wave does not originate at this lexically pre-assigned trough. If it did, the predicted stress pattern would be **atónicidade* *[ɐˈtɔː.ni.si.ðɐ.ðɨ], which is unattested. Instead, the actual stress pattern is [ˌɐ.tuˌni.siˈðaː.ðɨ], where the rhythmic wave clearly starts at the final syllable *-de*. The rhythmic wave theory, as presently formulated, does not specify under what conditions a lexically pre-assigned trough functions as the starting point of the rhythmic wave, and under what conditions it fails to do so. It is precisely the notion of PCE that makes these conditions explicit.

6. Stress clash avoidance (provisional)

When the metrification domain contains an **odd** number of syllables, to avoid a stress clash between a potential **word-initial unary foot** and the following **binary foot**, metrification is assumed to proceed as follows. Note that these generalisations are **highly provisional** and thus require further investigation.

6.1. Five or more syllables

Only the **leftmost foot** is **ternary**, or only the **initial syllable** is **unfooted**, while **subsequent feet** remain **binary trochees** (original data "ˌcomputaˈdor" and "comˌputaˈdor" — theoretically reinterpreted by the author — from Mateus et alii 2003: 1059):

computador /konput-a-dor-0/ → kon.pu.ta.do.ɽ → (ˌkon.pu.ta)(ˈdo.ɽ) → [ˌkõ.pu.tɐˈðoːɾ]
computador /konput-a-dor-0/ → kon.pu.ta.do.ɽ → kon(ˌpu.ta)(ˈdo.ɽ) → [kõˌpu.tɐˈðoːɾ]
computador /konput-a-dor-0/ → kon.pu.ta.do.ɽ → *(ˌkon)(ˌpu.ta)(ˈdo.ɽ) → *[ˌkõˌpu.tɐˈðoːɾ]

(Note that optional unary footing of the initial syllable, as in *armoricano* /armɔrik-an-o/ → ar.mɔ.ri.ka.no → (ˌar)(ˌmɔ.ri)(ˈka.no) → [ˌɐɾˌmu.ɾiˈkɐː.nu], is not excluded as a possible metrification — original data, metrical-gridded without footing and theoretically reinterpreted by the author, are from d'Andrade & Laks 1992: 23). From this, the following descriptive generalisation can be formulated: all syllables are parsed into binary feet, suggesting the constraints **FtBin(σ)** and **Parse(σ)**, except when the metrification domain contains an odd number of syllables, where only the leftmost foot is ternary or only the initial syllable is unfooted so as to avoid a stress clash, suggesting the ranking ***Clash ≫ FtBin(σ) = = Parse(σ)**.

6.2. Three syllables

The **initial** syllable is **unfooted**:

pequeno /peken-o/ → pe.ke.no → pe(ˈke.no) → [pɨˈkeː.nu]
pequeno /peken-o/ → pe.ke.no → *(ˌpe)(ˈke.no) → *[ˌpɨˈkeː.nu]
pequeno /peken-o/ → pe.ke.no → *(ˈpe.ke.no) → *[ˈpeː.kɨ.nu]

(Note that optional unary footing of the initial syllable, as in *doutores* /dout-or-0-s/ → doṷ.to.r0s → doṷ.to.res → (ˌdoṷ)(ˈto.res) → [ˌdoˈtoː.ɾɨʃ], is not excluded as another possible metrification.) The constraint ranking ***Clash ≫ FtBin(σ) = = Parse(σ)**, inferred in Section 6.1, does not evaluate the candidate *(ˈpe.ke.no) as a loser, and it is therefore necessary to introduce **HdFtBin(σ)** as a tie-breaking constraint.

7. The constraint set (provisional)

From the discussion in Sections 4–6, the following set of constraints and their rankings are inferred:

*CLASH ≫ FTBIN(σ) = = PARSE(σ);
TROCHEE(σ);
HDFTBIN(σ);
ALIGN-R(Ft, σ_{final}_PCE) ≫ ALIGN-R(Ft, PrWd);
ALIGN-R(HdFt, σ_{final}_PCE) ≫ ALIGN-R(HdFt, PrWd)

8. Problems of the rhythmic wave theory

As stated in the last paragraph of Section 2, Mateus & d'Andrade (2000) and Mateus *et al.* (2003) attempt to account for Portuguese stress pattern by analysing it in terms of a "rhythmic wave". This theory appears to face the following problems:

(i) The problem of trough–peak conflict: according to rhythmic wave theory, the syllable *-me-* in *nú**me**ro* should, in principle, be lexically specified as a pre-assigned trough, yet in *nu**mé**rico* the same *-me-* bears primary stress. A contradiction thus arises, in that a syllable specified as a trough also functions as a stress peak. By assuming PCE, however, such a contradiction can be avoided.

(ii) The insufficient account of the final unstressed syllable in proparoxytones: the rhythmic wave theory alone cannot explain why the final syllable *-ro* in *núme**ro*** is unstressed and ultimately requires the introduction of an additional principle independent of the rhythmic wave itself. By contrast, the metrical foot theory can account for this by means of a single principle: "a syllable that is not footed is unstressed."

(iii) The explanation of word-initial secondary stress: in words such as ***com**putadora*, where the metrification domain contains an odd number of syllables, the word-initial secondary stress may fall on a syllable corresponding to a trough under the rhythmic wave theory and thus requires secondary stress assignment by an additional principle. By contrast, under the metrical foot theory, it is naturally derived from the same principle of footing as other secondary stresses, on the basis of the constraint set introduced in Section 7.

Thus, whereas the rhythmic wave theory requires multiple independent principles to account for trough–peak conflicts, the final unstressed syllable in proparoxytones and word-initial secondary stress in odd-syllable metrification domains, the metrical foot theory, by postulating /0/ and PCE, derives primary stress, secondary stress and unstressed syllables through a single footing mechanism based on a single set of constraints. Accordingly, the analysis based on the latter theory appears more concise and theoretically more economical.

9. Plausibility of the postulation of /0/

9.1. Historical corroboration: from Latin to Portuguese

Comparing the Portuguese nouns *amador* /am-a-dor-**0**/ and *razão* /razon-**0**/ with their Latin etymons *amatorem* /amā-t-ōr-**e**-m/ and *ratiōnem* /rat-iōn-**e**-m/ shows that the Portuguese /0/ corresponds to the Latin segment /e/. Many other examples of this kind can also be found. This suggests that, in this case, the Portuguese **/0/ was established as the result of the Latin /e/ having its quality deleted while its quantity was preserved**, thereby **preventing main-stress displacement through the preservation of the head trochaic foot**.

9.2. Synchronic corroboration: plural in *-es* [-ɨʃ] as the /-0s/ realisation

We analyse plural nouns and adjectives ending in -***es***, such as *contador**es***, as /kont-a-dor-**0**-s/, with the thematic vowel /-0/. In the singular *contador* /kont-a-dor-**0**/, μ_0 is taken to be associated with the final rhotic /-r/ during syllabification, yielding (ˌkon.ta)(ˈdo.r̩) when metrified, which surfaces as [ˌkõ.tɐˈðoːɾ] through resyllabification. In the plural, on the other hand, the /0/ in the sequence /-C**0**s/# ultimately surfaces as [ɨ] through the linking of μ_0 to the epenthesised vocalic quality, yielding the surface form [ˌkõ.tɐˈðoː.ɾ**ɨ**ʃ] from the underlying representation /kont-a-dor-**0**-s/. This analysis avoids positing an underlying /e/ in the plural — as in /kont-a-dor-**e**-s/ — which would have to be deleted in the singular /kont-a-dor-**e**/ at the surface level. Such a move would be incompatible with the many lexical items that retain final /-e/ [-ɨ] — e.g. *árvor**e***, *ar**e***, *hectar**e***, *Zêzer**e***. Note finally that, by **positing /-0/ alongside /-a/, /-e/, /-o/ as a thematic vowel**, **all inflexional words** — namely, nouns, adjectives and verbs — can be analysed as **sharing the common structure [[[radical] thematic vowel] inflexional suffixes]**, while at the same time allowing **the stress pattern to be derived solely on the basis of phonological information.**

9.3. Regional EP corroboration: paragoge

In regional varieties, words ending in /-r**0**/ or /-l**0**/ may surface as [-ɾ**i**] ~ [-ɾ**ɨ**] or [-l**i**] ~ [-l**ɨ**]. For instance, the infinitive *comer* /kɔm-e-r**0**/ may surface as [kuˈm.eː.ɾ**i**] or [kuˈmeː.ɾ**ɨ**], with paragogic [i] or [ɨ], rather than as the standard [kuˈmeːɾ]. At the surface level, μ_0 is associated with [ɾ] in the standard variety, whereas in those Northern varieties **μ₀ may instead be associated with the epenthetic quality [i] or [ɨ]**, with **the stress pattern remaining unchanged**. In other words, the "ghost note" /0/ thus occasionally becomes associated with *inserted* vocalic timbre, thereby becoming a "full note" [i] or [ɨ], yet without altering the rhythm. This indicates that **μ₀, irrespective of its association with any quality — consonantal or vocalic — constitutes the weak beat in a strong–weak alternation** and that this unit is not a mere stray/floating mora, but an element that projects a syllable nucleus.

9.4. Cross-Romance corroboration: Portuguese vs. Spanish

The main stress position of the future subjunctive forms *trabalharmos* and *trabalhar* can be determined solely on the basis of phonological information, by analysing the **future subjunctive suffix *-r*** as the **PCE /-r0/ with /0/**:

trabalharmos /trabaʎ-a-**r0**-mos/ → tra.ba.ʎa.**r̩**.mos → (ˌtra.ba)(ˈʎa.**r̩**)mos
trabalhar /trabaʎ-a-**r0/** → tra.ba.ʎa.**r̩** → (ˌtra.ba)(ˈʎa.**r̩**)

This future subjunctive suffix is ***-re*** **/-re/ with /e/** in Spanish, which, like -r /-r0/ in Portuguese, functions as a PCE and renders the final syllable of the PrWd extrametrical:

trabajáremos /trabax-a-**re**-mos/ → tra.ba.xa.**re**.mos → (ˌtra.ba.)(ˈxa.**re**)mos
trabajare /trabax-a-**re/** → tra.ba.xa.**re** → (ˌtra.ba.)(ˈxa.**re**)

From this, it becomes evident that, **in stress computation, the Portuguese "ghost note" /0/ functions in exactly the same way as the Spanish "full note" /e/**. Furthermore, the stress patterns of the Portuguese noun *coração* and verb *falar* and their Spanish cognates *corazón* and *hablar*, for instance, can likewise be **derived in parallel by positing /0/**:

coração /korason-**0/** → ko.ra.so.n̩ → (ˌko.ra)(ˈso.n̩) → [ˌku.ɾɐˈsɐ̃ũ̯]
corazón /koraθon-**0/** → ko.ra.θo.n̩ → (ˌko.ra)(ˈθo.n̩) → [ˌko.ɾaˈθɔn]
falar /fal-a-r**0/** → fa.la.ɾ̩ → fa(ˈla.ɾ̩) → [fɐˈlaːɾ]
hablar /habl-a-r**0/** → a.bla.ɾ̩ → a(ˈbla.ɾ̩) → [aˈβlaɾ]

The difference between these two languages lies primarily in the degree of complexity of the phonological processes that apply after stress assignment, from the intermediate representations (ˌko.ra)(ˈso.n̩) and (ˌko.ra)(ˈθo.n̩), fa(ˈla.ɾ̩) and a(ˈbla.ɾ̩), respectively. Thus, **the model proposed**

in this paper may be considered extendable to Spanish as well.

9.5. Orthographic corroboration

The Portuguese orthographic system employs a diacritic in *túnel*, just as in *túnica*, even though it is superficially paroxytone, whereas it employs no diacritic in *tonel*, just as in *modelo*, even though it is superficially oxytone. The reason is that, once the /0/ is visualised, ***tú****.ne.l-0* becomes proparoxytone, much like ***tú****.ni.c-a*, while *to.****ne****.l-0* becomes paroxytone, much like *mo.****de****.l-o*. Without the native speakers' intuitive awareness of *-0*, the presence or absence of the diacritic cannot be accounted for in a principled manner.

9.6. Theoretical parsimony: economy of representation

Under the theoretical framework of this paper, the stress-pattern opposition between the quasi-minimal pair *tam****bém*** and ***lam****bem* can be easily explained by the final /0/ in *também* /tan-ben-**0**/, which is absent from *lambem* /lanb-e-n/:

também /tan-ben**0**/ → tan.be.n̩ → tan(ˈbe.n̩) → [tɐ̃ˈbɐ̃ȷ̃]
lambem /lanb-e-n/ → lan.ben → (ˈlan.ben) → [ˈlɐ̃ː.bɐ̃ȷ̃]

Accordingly, **the stress-pattern difference can be attributed solely to their segmental sequences, in terms of temporal profile**. Without positing /0/, by contrast, one would additionally have to assign a lexically specified accent to *também*, thereby requiring two types of information in the lexicon: segmental and lexical-accentual. Moreover, **the mere application of the Weight-to-Stress Principle to the surface forms also fails**, since in both *também* [tɐ̃ˈ**bɐ̃ȷ̃**] and *lambem* [ˈlɐ̃ː.**bɐ̃ȷ̃**] the final syllable [-bɐ̃ȷ̃] is a heavy syllable with two moras, so WSP cannot distinguish them. Therefore, **lexical specification would be required** once again in this case.

10. Marked metrification reflected in orthography

When **marked metrification** emerges due to a PCE, the stressed nucleus is **typically indicated orthographically** by an acute accent or a circumflex accent, since the outcome **deviates from the default stress position** — e.g. *âmbar*, *côdea*, *número*, *pêssego*, *sótão*, *túnel*, *académico*, *agradável*, *trabalhávamos*, *trabalháveis*. This close correlation suggests that **speakers' prosodic intuitions** about **stress location** are reflected in the standard orthography, thereby providing independent support for the analysis.

11. Conclusion

Thus, Portuguese stress can be fully accounted for by the **interaction of:**

- **Qualityless vowel /0/ bearing one mora**
- **Prosdy-Conditioned Extrametricalisers (PCEs)**
- **General metrical theory formalised through OT constraints**

without recourse to morphologically-driven stress assignment. The principles and parameter of primary stress assignment in Portuguese are as follows:

Principles

- Trochaic binary footing from the right edge
- The rightmost foot functions as the head foot

Parameter: determination of the right edge

- **Value 1:** final syllable — unmarked default
- **Value 2:** penultimate syllable — marked (when a PCE satisfies the condition)

In this way, Portuguese stress patterns are reduced to two possibilities in terms of primary stress placement, yielding a relatively simple system. However, these patterns are superimposed on a wide variety of morphological structures, and it is the attempt to describe stress assignment from the perspective of those morphological structures that appears to have contributed to the observed complexity.

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